

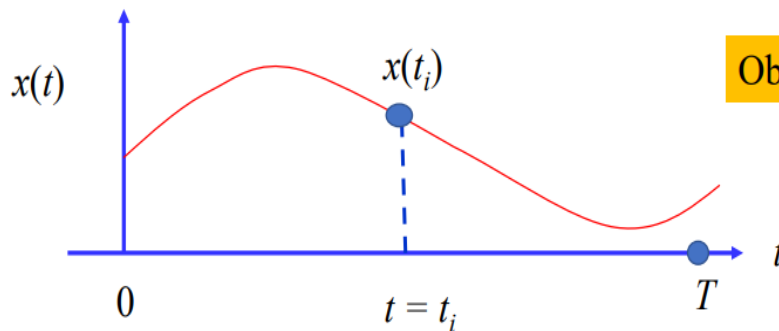
Data Sommmunication

1st Class

1. Define signal

Answer: 3 no slide 1st para with diagram

✓ Any **variable** that conveys information is called **signal**. If $x(t)$ is a time dependent variable, may be voltage, current or power of a network indicating changes in amplitude with time.



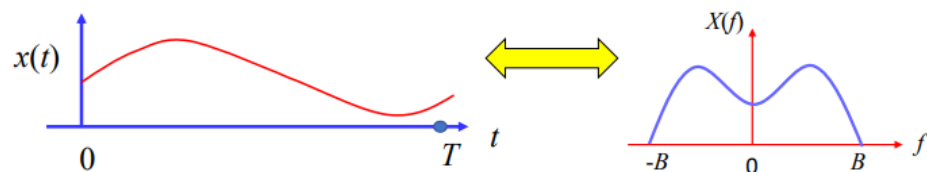
2. How signal is transform in frequency?

Answer: 4 no slide with diagram

For periodic signal

For non-periodic signal

✓ **Fourier series** or **Fourier transform** is the basic mathematical method of conversion of a signal from time domain to **frequency domain** and vice versa.

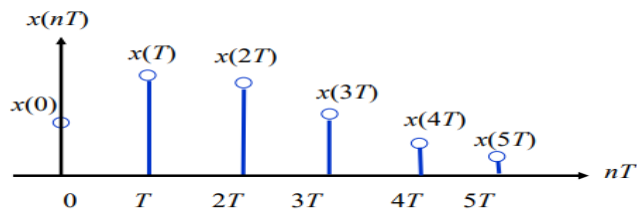


$$x(t) \leftrightarrow X(f)$$

Highest frequency of the signal

3. Compare discrete time signal and digital signal

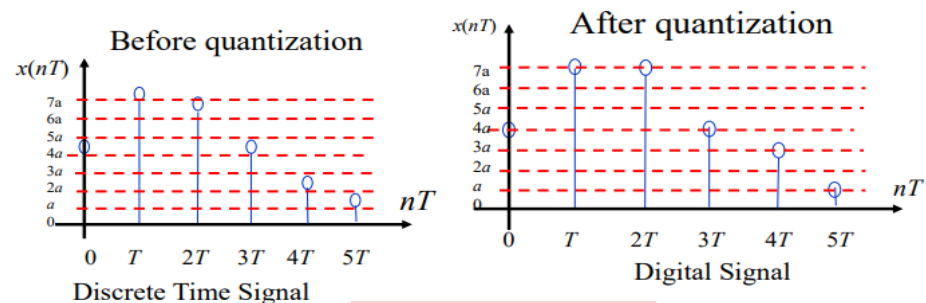
Answer: 8 no slide, 2nd para with diagram and slide 9 full



If the time axis of a signal is discrete but its amplitude axis remains continuous then it is called **discrete time signal**.

Digital signal

- ✓ If both time and amplitude axis of a signal is discrete, it is called digital signal.
- ✓ Amplitude axis is marked with several discrete levels with level spacing of a . If a sampled signal is quantized (amplitude of each sample is adjusted with nearest discrete level), it becomes **digital signal**.



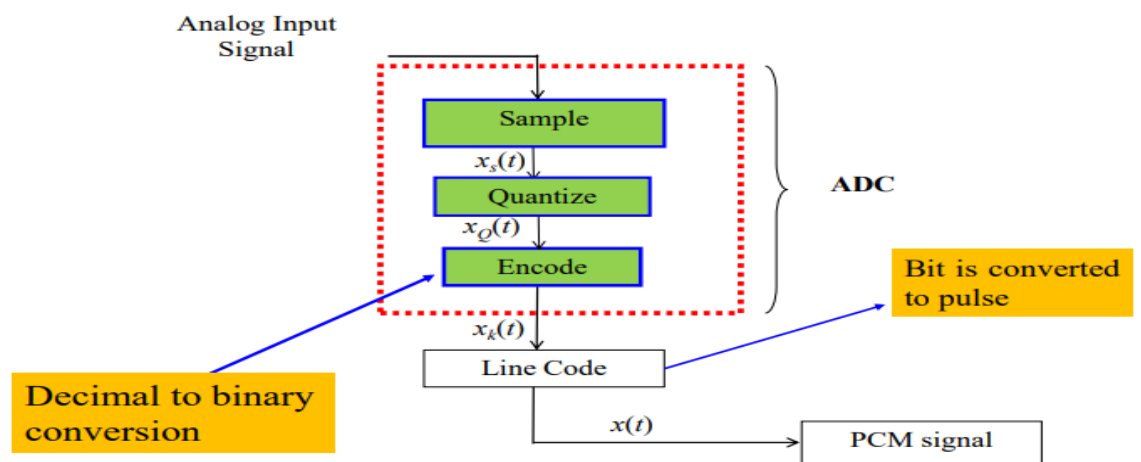
$T \rightarrow$ sampling period
 $a \rightarrow$ Level spacing

9

4. Draw the complete diagram of Analog to Digital Conversion

Answer: Slide 13.

Analog to Digital Conversion



5. State Nyquist sampling criteria?

Answer: Slide 17.

Nyquist sampling criteria:

The sampling frequency of a baseband signal should be at least twice of its highest frequency to recover the baseband signal from the sampled signal.

Nyquist Sampling Criteria (সহজ ভাষায়)

কথাটা কী?

কোনো continuous (একটানা) সিগন্যালকে যদি আমরা ডিজিটালভাবে নিতে চাই (sampling করি), তাহলে sampling frequency এমন হতে হবে যেন আসল সিগন্যালটা আবার ঠিকভাবে ফেরত পাওয়া যায়।

If we want to take a continuous signal digitally (i.e., sample it), then the sampling frequency must be such that the original signal can be perfectly reconstructed again.

6. Draw the Spectrum

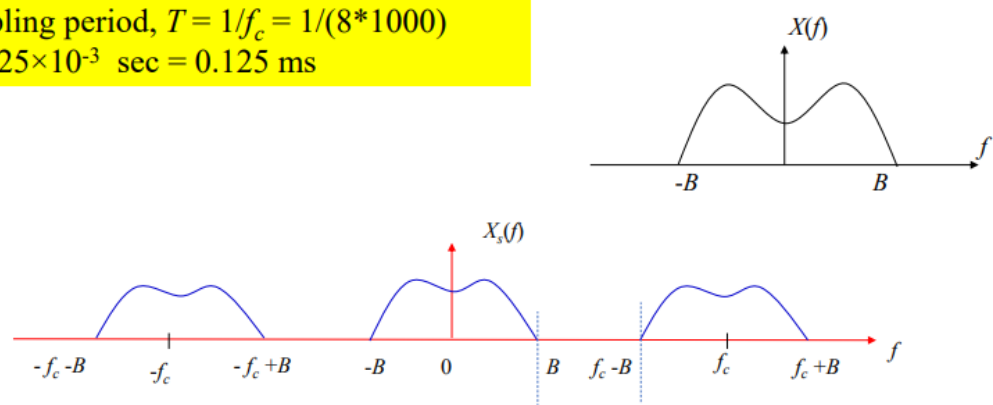
Answer: Example slide no 18

Example:

An analog signal of highest frequency of $B = 3.4$ KHz sampled maintaining a guard band of 1200Hz, find sampling rate f_c and sampling period T .

The sampling rate (or frequency), $f_c = G.B + 2B = 1.2 + 2 \times 3.4 = 8$ KHz

Sampling period, $T = 1/f_c = 1/(8 \times 1000)$
 $= 0.125 \times 10^{-3}$ sec = 0.125 ms



18

2nd Class

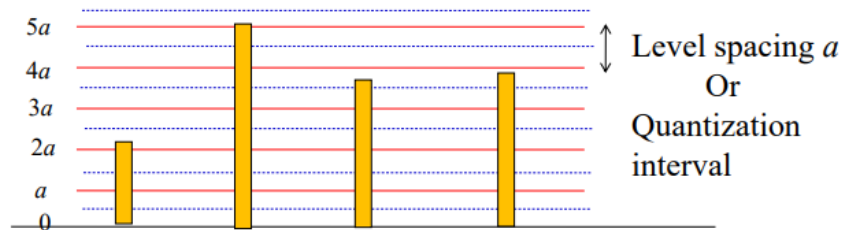
7. How quantization is done and what its impact on received signal

Answer: Slide 19 full with diagram.

2. Quantization of Sampled Signal

✓ Second step of digitations of an analog signal is **quantization** (the amplitude of a sample is adjusted with its nearest discrete level) of sampled signal into some known discrete amplitude levels.

✓ Due to quantization the signal losses its originality and an error is introduced with the signal known as **quantization error**.



How quantization is done and what its impact on received signal?

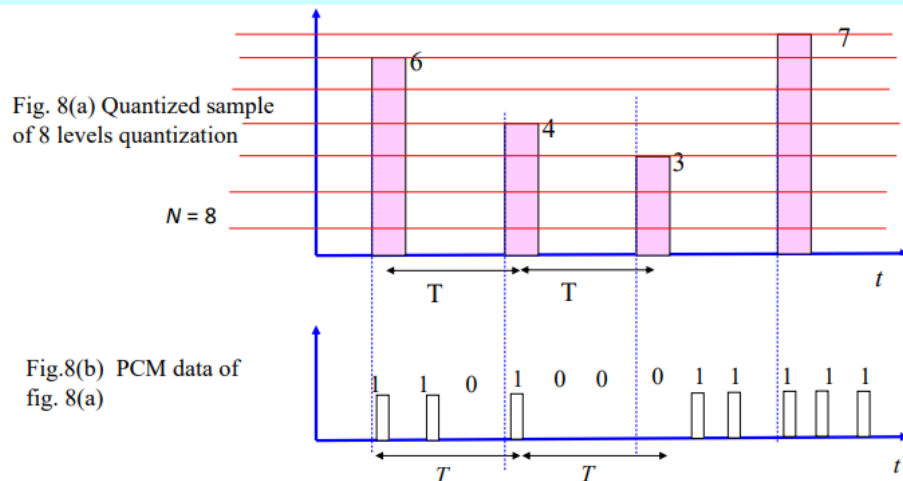
19

8. Show PCM technique with diagram.

Answer: 20 no slide with diagram.

3. Pulse Code Modulation

✓ Pulse Code Modulation (PCM) is essentially final step of analog to digital conversion where each quantized sample is converted to equivalent binary pulses, resembles to decimal to binary conversion.



Show PCM technique with diagram.

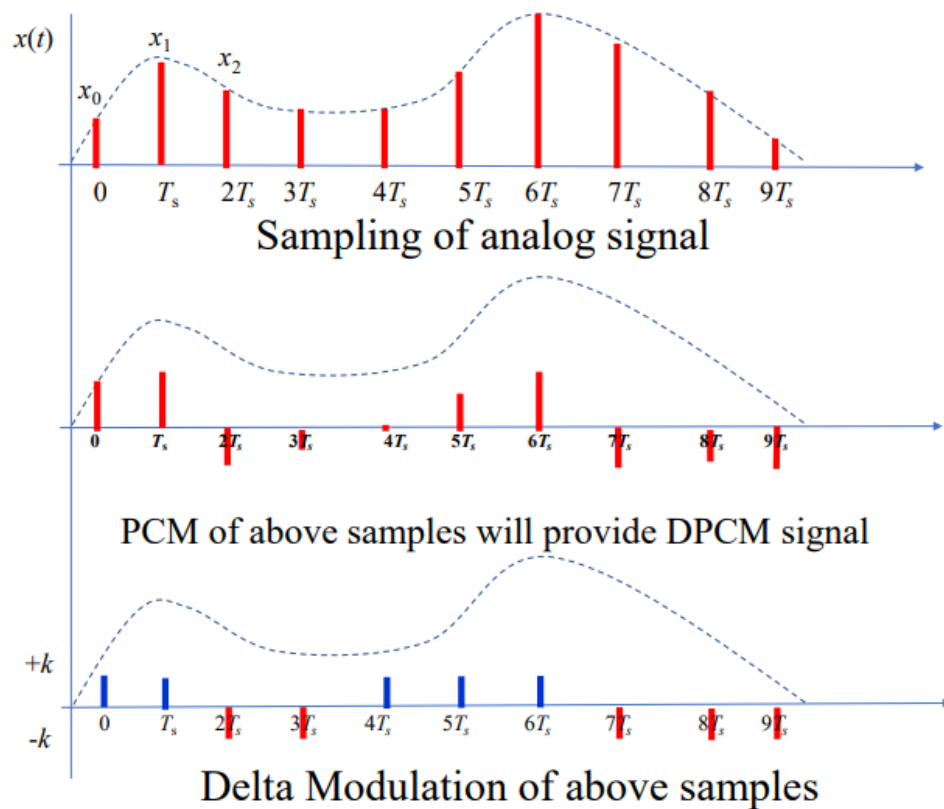
20

9. Give the concept of DPCM and Delta modulation with diagram?

Answer: Slide 25 and 26 with diagram.

Differential Pulse Code Modulation (DPCM)

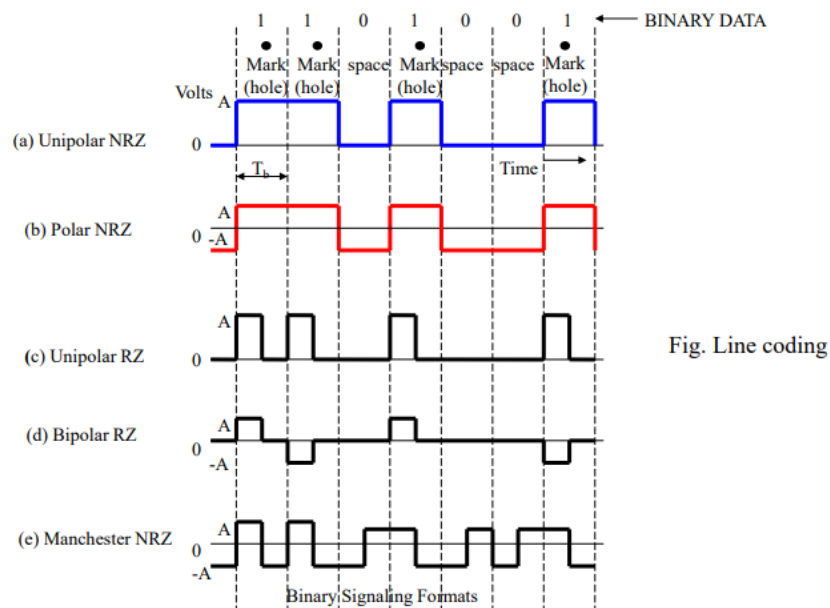
- ✓ It is found that average amplitude of sample to sample difference is less than that of original sampled wave of voice signal therefore less number bit is required for PCM, maintaining same SQR.
- ✓ If PCM is done on consecutive pulse difference instead of individual pulse known as **Differential Pulse Code Modulation** (DPCM). When sample to sample difference is expressed by a single bit ($+k$ or $-k$) then the modulation scheme is called **delta modulation** is considered as an special case of DPCM.



10. Problem on line coding.

Answer: Slide 31 but polar, unipolar, Bi polar egulor definition asbe na just bujhar jonno dewa ache.

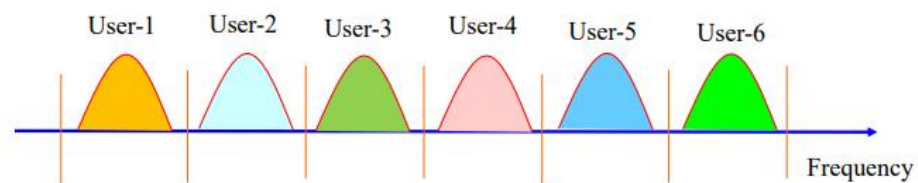
According to above classification different types of line coding is shown in fig. below.



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11. Define multiplexing and multiple access?

Answer: Slide 41 with diagram.



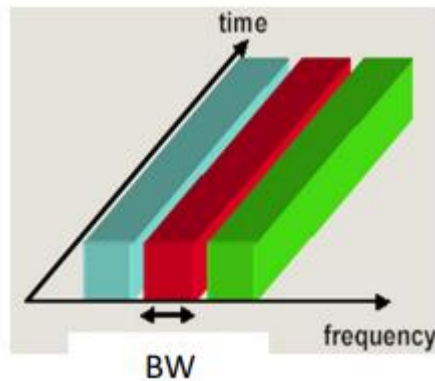
Each BW is a **physical channel** can accommodate a signal from a user

Multiplexing is process to combine **multiple** signal to transmit through a single channel or media.

Multiple access means many (several terminals) can **access** at on the same transmission medium, where each user/node/terminal shares the medium in the form of physical channel (TS, BW, code etc.). **Multiple access is the application of multiplexing.**

12. Give the concept of Frequency division multiplexing(FDM).

Answer: Slide 43 1st diagram with 1st para.

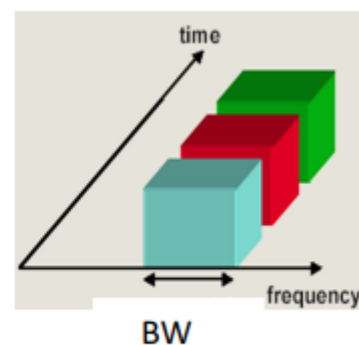


- ✓ In frequency-division multiple access (FDMA), the available bandwidth is divided into frequency bands. Each station is allocated a band to send its data. In other words, each band is reserved for a specific station, and it belongs to the station all the time.

13. Give the concept of Time division multiplexing (TDM).

Answer: 1st diagram with 1st para with yellow color.

In time-division multiple access (TDMA), the stations share the bandwidth of the channel in time. Each station is allocated a time slot during which it can send data.



Computer Network (2. Lec 2 625P.pdf)

14. What is the function of modem in Computer Network?

Answer: 1st Slide, 2nd para.

MODEM is connected between PC and Internet cable to convert the rectangular data pulse from the computer into continuous wave (called **modulation** during transmission) . During reception (for example downloading of data) the modulated wave is converted to digital pulse train called **de-modulation**.

15. Why Digital modulation is necessary in wired communication?

Answer: 2nd slide, para 1,2

- ❖ To reduce the bandwidth of the baseband signal according to the capacity of the transmission medium and receiver circuits.
- ❖ Multiple signal can be transmitted based on FDMA.
- ❖ Baseband signal-এর bandwidth কমিয়ে transmission medium এবং receiver circuit-এর ক্ষমতার সাথে মিলিয়ে নিতে হয়।
- ❖ FDMA-এর ভিত্তিতে একসাথে একাধিক signal পাঠানো যায়।

16. Why Digital modulation is necessary in wireless communication?

Answer: 3rd slide with any 3 para of all.

❖ In wireless communication digital modulation is necessary to avoid interferences of surrounding users with choose of appropriate carrier. (উপযুক্ত carrier frequency নির্বাচন করে আশেপাশের user-দের interference এড়ানো যায়।)

❖ Noisy immunity is increased for a high frequency modulated wave. (উচ্চ frequency-এর modulated wave হলে noise-এর প্রভাব কম হয় (noisy immunity বাড়ে)।)

❖ Size of an antenna depends on wavelength of a signal. After modulation (using high frequency carrier) wavelength of the modulated wave is reduced hence size of the antenna is reduced. (Antenna-এর আকার signal-এর wavelength-এর উপর নির্ভর করে। Modulation (high frequency carrier ব্যবহার করে) করার ফলে wavelength কমে যায়, তাই antenna ছোট করা যায়।)

❖ A modulated wave can propagate far longer way than the baseband signal. (Modulated wave, baseband signal-এর তুলনায় অনেক দূর পর্যন্ত যেতে পারে।)

1. Write basic principle of BFSK.

Ans: Slide 4 or unorder shadow color not block diagram

Binary Frequency Shift Keying (BFSK)

In binary FSK two sinusoidal waves of frequency f_1 and f_2 over a period of T or $[kT, (k+1)T]$ are used to represent binary logic 1 and 0 respectively. The waves are like,

$$S_1(t) = A \cos(2\pi f_1 t + \theta) \quad ; \quad kT \leq t \leq (k+1)T \quad \text{for logic 1}$$

$$S_2(t) = A \cos(2\pi f_2 t + \theta) \quad ; \quad kT \leq t \leq (k+1)T \quad \text{for logic 0}$$

; where $f_1 > f_2$ and θ is the initial phase of sinusoidal wave i.e. at the starting point symbol wave.

2. Draw BFSK modulator circuit.

Ans: Slide 5 Only figure 1 diagram not description.

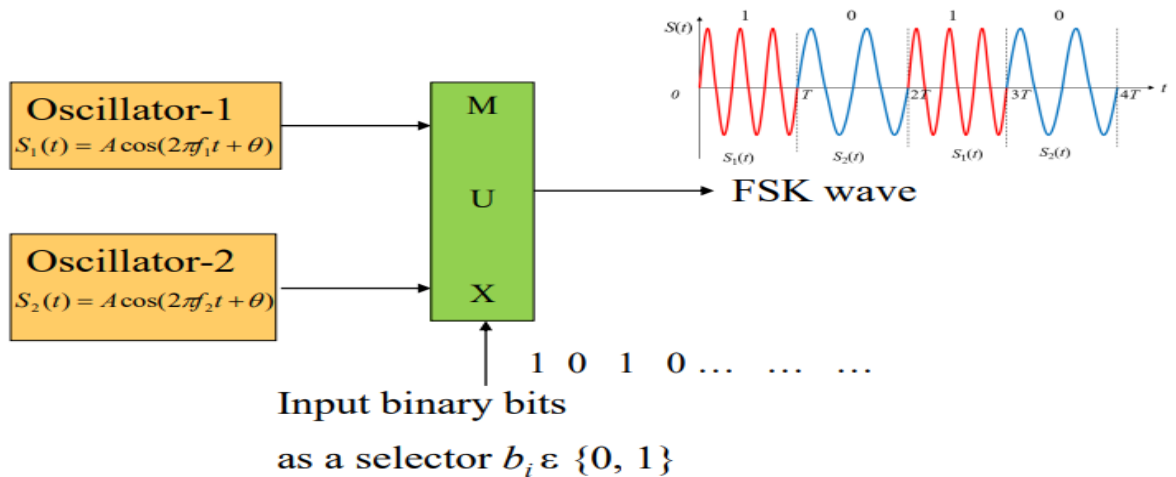
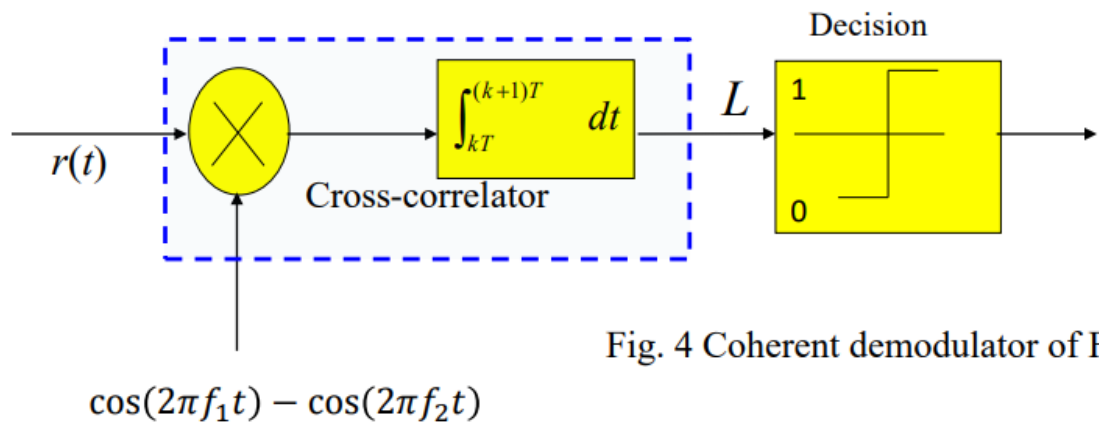


Fig. 1 Binary FSK modulator

3. Draw Coherent demodulator circuit of BFSK.

Ans: Slide 6 Only diagram and with $\cos(2\pi f_1 t) - \cos(2\pi f_2 t)$



4. Give basic principle of BPSK.

Ans: Slide 8 purotai

Binary Phase Shift Keying (BPSK)

In BPSK binary logic 1 and 0 are presented by two sinusoidal waves of same frequency and same peak amplitude over the period $[kT, (k+1)T]$ but initial phases are 0 and π respectively. BPSK wave is expressed like,

$$S_1(t) = A \cos(2\pi f_c t + 0) = A \cos(2\pi f_c t) ; kT \leq t \leq (k+1)T \quad \text{for logic 1}$$

$$S_2(t) = A \cos(2\pi f_c t + \pi) = -A \cos(2\pi f_c t) ; kT \leq t \leq (k+1)T \quad \text{for logic 0}$$

Wave shape of BPSK is shown in fig.5.

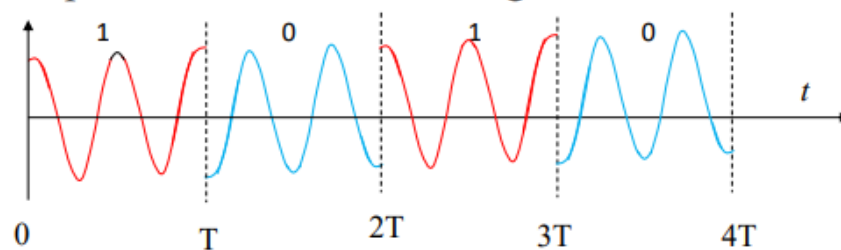


Fig. 5 BPSK waveform

5. Why M-ary PSK is preferable compared to BPSK?

Ans: Slide 11 er uporer shadow color only

M-ARY PSK

In BPSK bit rate and modulation symbol rate are equal. In MPSK $n = \log_2 M$ bits are represented by a modulation symbol hence bit rate is n times higher than symbol rate. Therefore n times more information can be transmitted by MPSK (for the same **baud** or **symbol** rate) scheme compared to that of BPSK at the expense of bit error rate. MPSK wave is expressed as,

In BPSK bit rate and modulation symbol rate are equal. In MPSK $n = \log_2 M$ bits are represented by a modulation symbol hence bit rate is n times higher than symbol rate. Therefore n times more information can be transmitted by MPSK scheme compared to that of BPSK

6. Derive the initial phase angles of 8-PSK.

Ans: Slide 11 er nicher 2 line only. In 8 PSK evabe ans start korbo.

For example 8-PSK, $M = 8$ and $\theta_i = \frac{(2i-1)\pi}{8}$.

$$\text{Therefore, } \theta = \left\{ \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8} \right\}$$

7. Why Quadrature amplitude modulation is widely used in data communication? Give the expression of modulated wave?

Ans: Slide 18 er shadow lekha and first equation only.

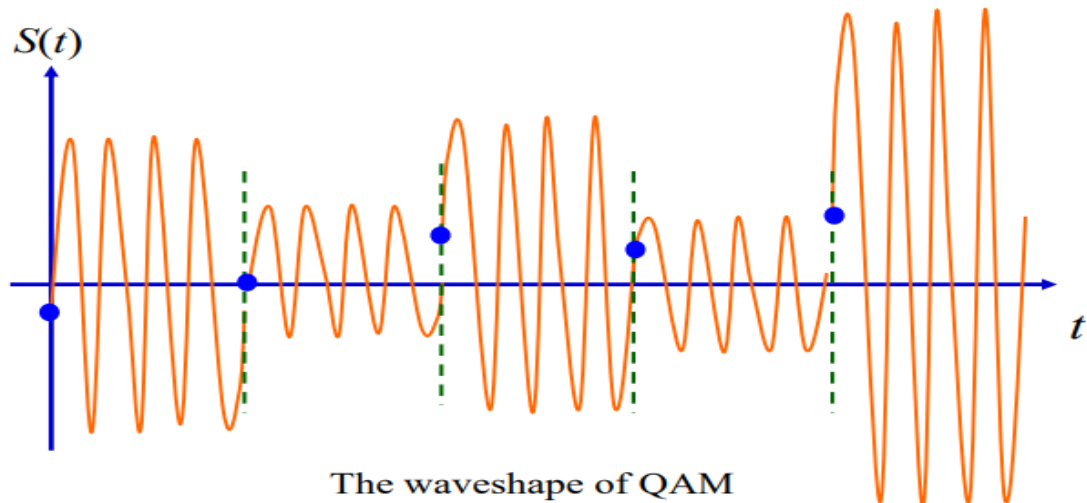
In QAM both amplitude and phase of the carrier are modulated to enhance bandwidth efficiency. For M-QAM the i^{th} symbol wave is expressed as,

$$S_i(t) = A_i \cos(2\pi f_c t + \theta_i) \quad ; i = 1, 2, 3 \dots M$$

(QAM-এ carrier-এর amplitude এবং phase দুটোই পরিবর্তন (modulate) করা হয়, যাতে bandwidth বেশি কার্যকরভাবে ব্যবহার করা যায়।)

8. Draw a simplified waves/waveshaps

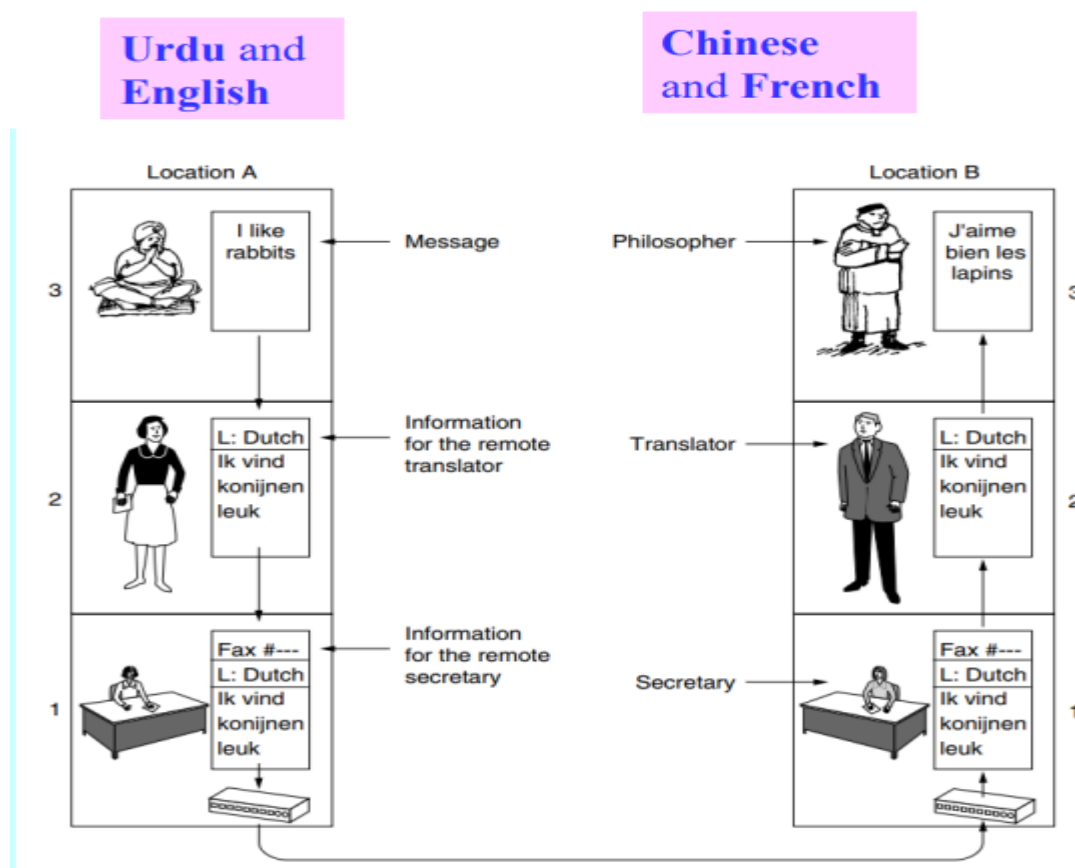
Ans: Slide 19 diagram.



4rth Class (Lec 3 625P.pdf)

1. Draw three layer architecture of two philosopher communication

Ans: Slide 18 er . (Only figure)



2. Mention two basic functions of Physical Layer.

Ans: Slide 28 er 2nd and 3rd point

☞ The physical form (e.g., voltages, frequencies, timing) in which data

bits (binary values 0 and 1) are represented. Actually it deals with shape of raw digital pulse i.e. types of line coding.

৩ The type of modulation (ASK, FSK, PSK, QPSK, 16-QAM etc.) to be used for transmitting digital data over analog transmission lines.

3. **Mention three basic functions of LLC (Logical Link Control).**

Ans: Slide 30 er nicher 3rd point only and 31 no slide er 2nd para and 32 slide er 1st para.

Frames are constructed from data string by adding special bit patterns to the beginning and end of each segment. This allows the receiving end to detect where each frame begins and where it ends.

Data string থেকে frame তৈরি করা হয়, যেখানে প্রতিটি segment-এর শুরু এবং শেষে বিশেষ bit pattern যোগ করা হয়।

এর ফলে receiver সহজেই বুঝতে পারে—

কোন জায়গা থেকে frame শুরু হয়েছে এবং কোথায় শেষ হয়েছে।

When a frame arrives and is corrupted or is lost for any reason in the network, it is retransmitted. The receiver confirms correct receipt of each frame by sending back an acknowledgement frame.

খন কোনো frame নেটওয়ার্কে এসে নষ্ট (corrupted) হয়ে যায় বা হারিয়ে যায়, তখন সেটি আবার পাঠানো হয় (retransmit করা হয়)।

Receiver প্রতিটি frame ঠিকভাবে পাওয়ার পর একটি acknowledgement (স্বীকৃতি) frame পাঠিয়ে নিশ্চিত করে যে সেটি সঠিকভাবে গ্রহণ করা হয়েছে।

Flow control: In general, not all communication devices in a network operate at the same speed. Flow control provides a means of avoiding a slow receiver from being swamped by data from a fast transmitter using some buffer.

Flow control: নেটওয়ার্কে সব ডিভাইস একই গতিতে কাজ করে না।

Flow control এমন একটি পদ্ধতি, যা দ্রুত sender থেকে আসা বেশি data দিয়ে ধীর receiver-কে overwhelmed (চাপের মধ্যে পড়ে যাওয়া) হওয়া থেকে রক্ষা করে, সাধারণত একটি buffer ব্যবহার করে।

4. **Compare error detection and error correction.**

Ans: Slide 31 er 2nd and 3rd para.

Error detection: When a frame arrives and is corrupted or is lost for any reason in the network, it is retransmitted. The receiver confirms correct receipt of each frame by sending back an acknowledgement frame.

Error correction: In case of multimedia traffic (image, video, voice etc.) the receiver corrects the frame used some complex algorithm like convolutional encoding, Linear block code, R-S code etc.

5. **Compare physical and logical address.**

Ans: Slide 36 er 1st 2 para.

The physical addressing implemented by the data link layer handles the addressing problem locally (within the LAN). MAC Address ensure that physical address of the computer is unique.

If a packet passes the network boundary (outside of LAN), we need another addressing system to help distinguish the source and destination systems (area wise addressing). IP Address is a logical address of the computer and is used to uniquely locate computer connected via a network.

5th Class(Lec 4 625P.pdf)

1. **What is the function of headers and trailers of a frame?**

Ans: Slide 3 er 1st para.

✓ The **header** of a frame, normally carries the source and destination addresses and other control information, and the **trailer**, which carries error detection or error correction redundant bits.

2. **Define frame with diagram.**

Slide 2 er diagram er uporer ongsho and diagram.

✓ Framing refers to the process of partitioning bit stream into discrete units or **blocks of data** called frame. Framing enables sending and receiving machines to **synchronize** the transmission and reception of data because frames have **detectable boundaries**.

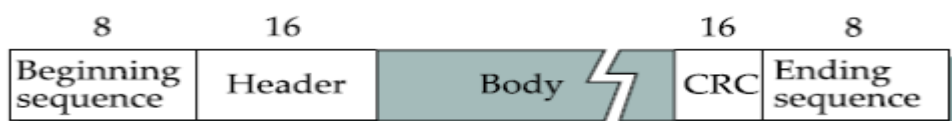
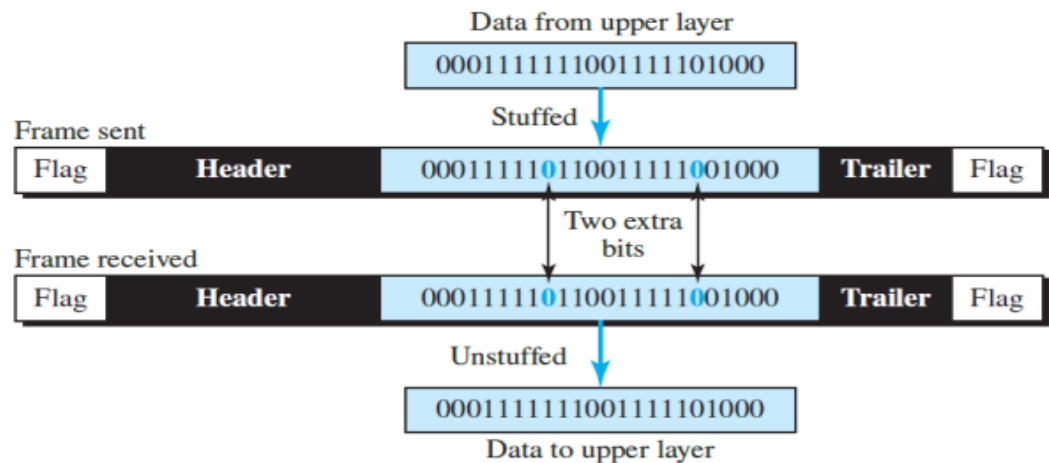


Fig.1 Frame format

3. **Show bit stuffing with appropriate diagram.**

Ans: Slide 10 er diagram only.

Fig. below gives an example of bit stuffing.



4. Compare bit and byte stuffing.

Ans: 11 slide all.

- ✓ With both bit and byte stuffing, a side effect is that the length of a frame now depends on the contents of the data it carries.
- ✓ For instance, if there are no **flag bytes** in the data, 100 bytes might be carried in a frame of roughly 100 bytes (best case).
- ✓ If, however, the data consists solely of flag bytes, each flag byte will be escaped and the frame will become roughly 200 bytes long (worst case). (**byte stuffing case**)
- ✓ With bit stuffing, the increase would be roughly 12.5% as 1 bit is added to every byte. (**bit stuffing case**)

5. Give the idea of error correction.

OR

Give the idea of 'error correction through retransmission' and 'autonomous error correction',

Ans: Slide 13 er 2nd and 3rd para

❑ The first method, *error correction through retransmission*, involves providing enough information in the data stream so the receiving node can detect an error. Once an error is detected the receiving node can then request the sender to retransmit that unit data.

❑ The second method is *autonomous error correction*, involves proving redundant information in the data stream so the destination node can both detect and correct any error autonomously.

6. **Write the graph of CRC algorithm.**

OR

Write the steps of the Cyclic Redundancy Check (CRC) algorithm.

Ans: Slide 17 er green color.

We can derive $T(x)$ from $M(x)$ in the following way:

1. Multiply the message polynomial $M(x)$ by x^r , which is an equivalent polynomial of a sequence derived by adding r zeros at the end of the message.
2. Divide $x^r M(x)$ by $G(x)$ and find the remainder $R(x)$.
3. Subtract the remainder $R(x)$ from the dividend $x^r M(x)$ to get $T(x)$.

It is obvious that $T(x)$ is exactly divisible by $G(x)$.

7. **Example-1**

Ans: Slide 18 and 19.

Example-1:

Given, Message bit string $M = 1\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0\ 1$ (10 bits)

Generator bit string $G = 1\ 1\ 0\ 1\ 0\ 1$ (6 bits)

Determine **remainder polynomial** $R(x)$ and **transmitted polynomial** $T(x)$.

Ans.

Given, Message bit string, $M = 1\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0\ 1$ (10 bits)

Generator bit string, $G = 1\ 1\ 0\ 1\ 0\ 1$ (6 bits)

Determine $M(x)$ and $G(x)$.

$$\therefore M(x) = x^9 + x^7 + x^3 + x^2 + x^0 \quad G(x) = x^5 + x^4 + x^2 + x^0$$

Degree of $G(x)$, $r = 5$.

Let us create another polynomial, $x^5 M(x) = x^{14} + x^{12} + x^8 + x^7 + x^5$

$$\begin{array}{r}
 x^5 + x^4 + x^2 + 1 \overline{) x^{14} + x^{12} + x^8 + x^7 + x^5} \\
 \underline{x^{14} + x^{13} + x^{11} + x^9} \\
 x^{13} + x^{12} + x^{11} + x^9 + x^8 + x^7 + x^5 \\
 \underline{x^{13} + x^{12} + x^{10} + x^8} \\
 x^9 + x^8 + x^6 + x^4 + x^2 + x^5 \\
 \hline
 G(x) \overline{) x^5 M(x)} \begin{array}{l} Q(x) \\ R(x) \end{array} \\
 \hline
 \begin{array}{r}
 12 \quad \begin{array}{|c|c|} \hline 79 \\ 72 \\ \hline 7 \end{array} \quad 6 \\
 \hline
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 x^{11} + x^{10} + x^8 + x^6 \\
 \underline{x^{11} + x^{10} + x^8 + x^6} \\
 x^9 + x^8 + x^7 + x^6 + x^5 \\
 \underline{x^9 + x^8 + x^6 + x^4} \\
 x^7 + x^5 + x^4 \\
 \underline{x^7 + x^6 + x^4 + x^2} \\
 x^6 + x^5 + x^2 \\
 \underline{x^6 + x^5 + x^3 + x} \\
 x^3 + x^2 + x
 \end{array}$$

Modulo 2 operation

$$\begin{aligned}
 \therefore R(x) &= x^3 + x^2 + x \leftrightarrow 0\ 1\ 1\ 1\ 0 \\
 \therefore T(x) &= x^5 M(x) - R(x) \\
 &= x^{14} + x^{12} + x^8 + x^7 + x^5 + x^3 + x^2 + x
 \end{aligned}$$

19

8. **How to avoid single bit error, two bit error and odd number of bit error in CRC?**

Ans: Slide 24 and 25 1,2,3

1. One common type of error is a **single-bit error**, which can be expressed as $E(x) = x^i$ when it affects bit position i . If we select $G(x)$ such that the first and the last term are nonzero, then we already have a two-term polynomial that cannot divide evenly into the one term $E(x)$.

2. For **two bit error**, $E(x) = x^m + x^r$; $m > r$

$$= x^r(x^{m-r} + 1) = x^r(x^k + 1)$$

In this case $G(x)$ has to be chosen such that it does not divide $x^k + 1$. For example if $G(x) = x^{15} + x^{14} + 1$ then $G(x)$ is unable to divide $x^k + 1$ for $k \leq 32,768$.

3. For **odd number of bits in error** then $E(x)$ will contains odd number of terms. For 3 bit error, $E(x) = x^5 + x^2 + 1$. If any polynomial has odd number of terms then it does not have any factor like $(x+1)$. If $G(x)$ has a factor of $(x+1)$ it would be unable to divide $E(x)$.

9. What will be the probability of $ex=gx$ **OR** what will happen $ex = gx$

OR

What is the probability of CRC failing to detect error?

Ans: Slide 26 er 5 no point.

5. If the error sequence and generator are identical then the CRC code will failed to detect error. The probability of such case,

$$P = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \dots (r+1) \text{ th term}$$

$= (1/2)^{r+1}$, which is very small

$$E(x) = G(x)$$

$$\begin{aligned} G &= 1\ 0\ 0\ 1\ 1\ 0\ 1\ 0\ 0\ 1 \\ E &= 1\ 0\ 0\ 1\ 1\ 0\ 1\ 0\ 0\ 1 \end{aligned}$$

26

Class Test-2 Class 3 to Class 5
porjonto

6th Class (Lec-4)

1. **Define flow control of LLC sublayer.**

Ans: Slide 28 er 3rd para.

- ✓ Flow control refers to the **regulating of the rate of data flow** from one device to another so that the receiver has enough time to consume the data in its receive buffer, before it overflows.

2. **Give basic working principle of Stop and Wait flow control.**

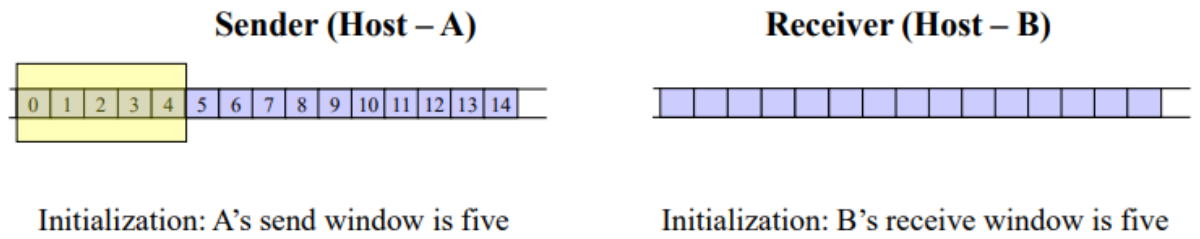
Ans: Slide 29

- ✓ The simplest form of flow control, known as **stop-and-wait** flow control.
- ✓ A source entity transmits a frame. After the destination entity receives the frame, it indicates its willingness to accept another frame by sending back an **acknowledgment** to the frame just received.
- ✓ The source must wait until it receives the **acknowledgment** before sending the next frame.
- ✓ The destination can thus stop the flow of data simply by withholding acknowledgment.

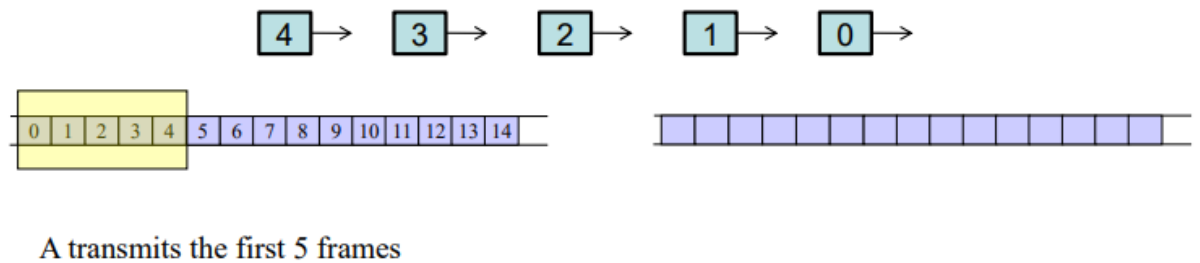
3. **Draw few steps of Sliding Window flow control.**

Ans: Slide 36,37 and 38 only diagram no description.

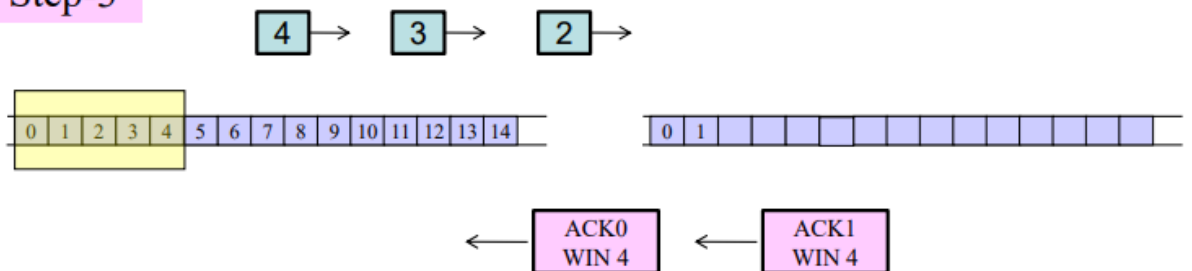
Step-1



Step-2



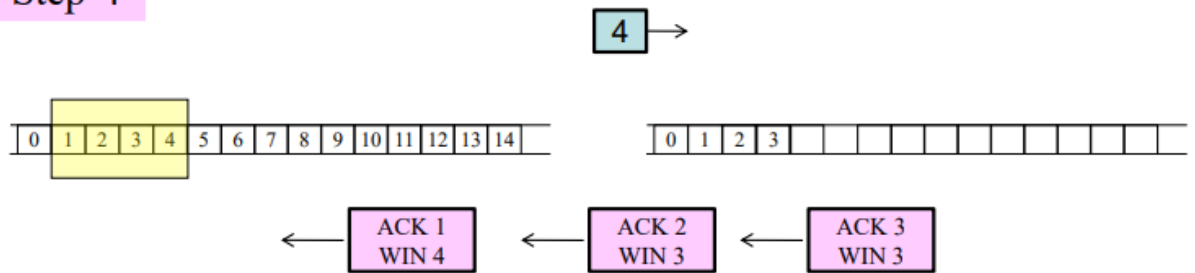
Step-3



A has not received any ACK so it keeps the window size 5 like before

B Receives frame 0 and 1 and sends the corresponding ACK. Each ACK carries a window size of 4. Thus once A receives the ACK for frame 0, it must change its window size to 4.

Step-4



A has ACK of frame 0 and advances the left side of its window by one. However, frame 0's ACK had the window size 4 of 4 therefore A cannot advance the right side of its window.

B Receives frame 2 and 3 and sends the corresponding ACK. Each ACK carries a window size of 3. Thus once A receives the ACK, it must change its window size to 3.

4. What is the main problem of random access protocol?

Ans: Slide 46 er seser parar “The efficiency theke of collision porjonto”

The efficiency of such protocol is very poor under heavy traffic condition because of frequent collision of packets. Under light traffic condition the throughput of such protocol is high due to low probability of collision

5. What is the main drawback of controlled access protocol?

Ans: 47 Slide 2nd para

Such protocol works efficiently during heavy traffic condition but during low traffic condition a station has to wait unnecessary for its turn. Token ring and token bus network use such protocol.

6. Compare exhaustive service and gated service of token ring protocol.

Ans: Slide 49, 50 - (a) & (b)

(a) The server serves all the packets (including those received during the transmission) in the queue (no packets of a customer is left) called exhaustive service.

(সার্ভার কিউতে থাকা সব প্যাকেট ট্রান্সমিশনের সময় যেগুলো আসে সেগুলোসহ) সম্পূর্ণভাবে সার্ভ করে—অর্থাৎ কোনো গ্রাহকের কোনো প্যাকেট বাকি থাকে না। একে **exhaustive service** বলা হয়।)

(b) The server serves only its queued packet which arrived just before capturing the token. In other words, the token is made free after transmitting the packets which were waiting before the transmission began i.e. the packets in the queue arrived after getting the token will not be served. This type service is called gated service.

(সার্ভার শুধুমাত্র সেই প্যাকেটগুলো সার্ভ করে যেগুলো টোকেন পাওয়ার ঠিক আগে কিউতে ছিল। অন্যভাবে বললে, ট্রান্সমিশন শুরু হওয়ার আগে যে প্যাকেটগুলো অপেক্ষা করছিল, সেগুলো পাঠানো শেষ হলেই টোকেন আবার মুক্ত করে দেওয়া হয়।)

অর্থাৎ টোকেন পাওয়ার পর নতুন করে যে প্যাকেটগুলো কিউতে আসে, সেগুলো সার্ভ করা হয় না। এই ধরনের সার্ভিসকে **gated service** বলা হয়।)

7. **Why CSMA/CA is used in wireless network?**

Ans: 53 slide 1st para.

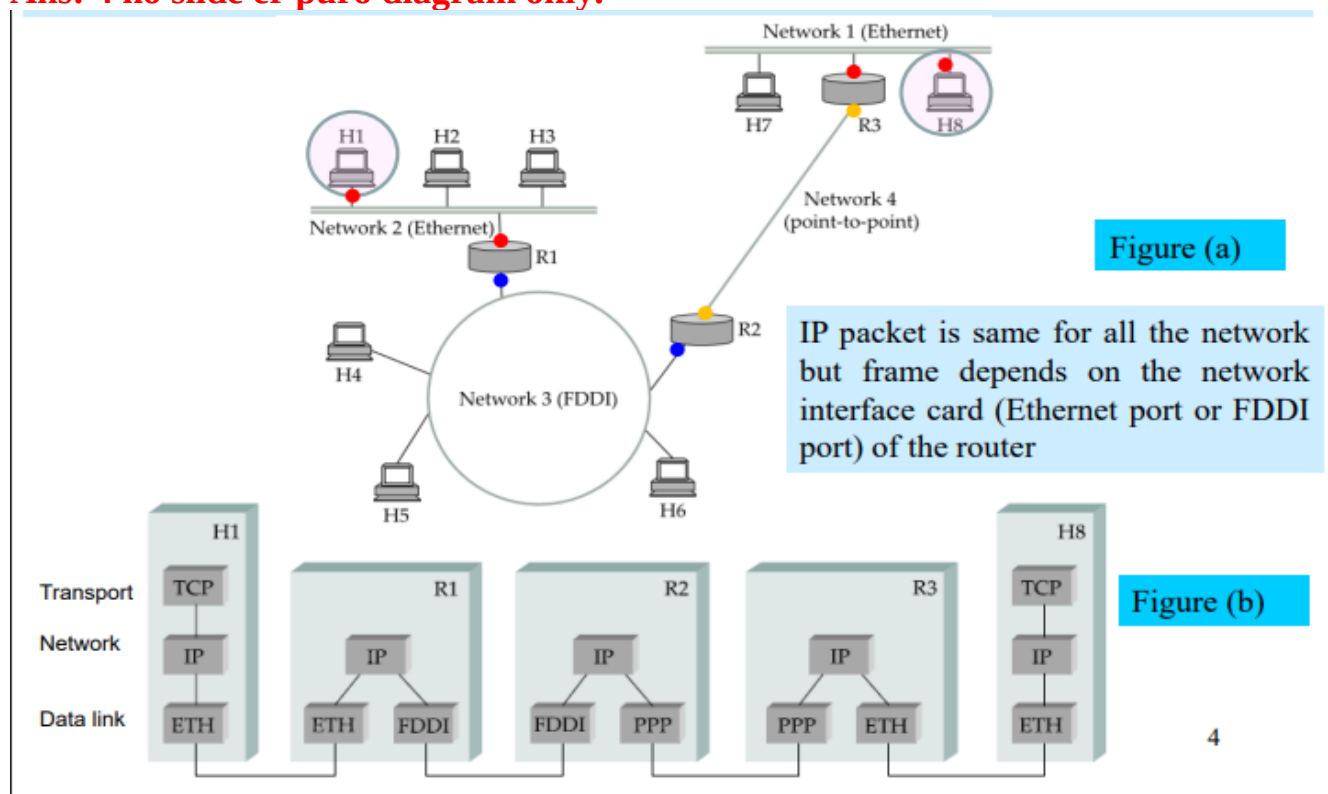
In wired communication both the transmitted and received signals are equally strong therefore collision can be detected easily, whereas situation is not favorable in wireless link, there collision avoidance (CA) is used in wireless network instead of collision detection (CD).

Class 7th and 8th

Class 8th Exam chilo tai only class 7th er question gulo likhe dicilo. (Lec-5)

1. Draw a network with layered architecture which is heterogeneous at data link layer but homogeneous at network layer.

Ans: 4 no slide er puro diagram only.



2. Mention the function differentiated service of IP packet.

Ans: Slide no 7 decription only.

- ❖ It is intended to distinguish between different classes of service. Various combinations of **reliability** and **speed** are possible.
- ❖ For **digitized voice**, fast delivery is more essential than accurate delivery.
- ❖ For **file transfer**, error-free transmission is more important than fast transmission.
- ❖ This field is related to three parameters: {Delay, Throughput, Reliability} in communication.

৳ এটি বিভিন্ন ধরনের সার্ভিসের শ্রেণি (class) আলাদা করার জন্য ব্যবহৃত হয়। বিভিন্ন ধরনের নির্ভরযোগ্যতা (reliability) এবং গতি (speed)-এর সমন্বয় সম্ভব।

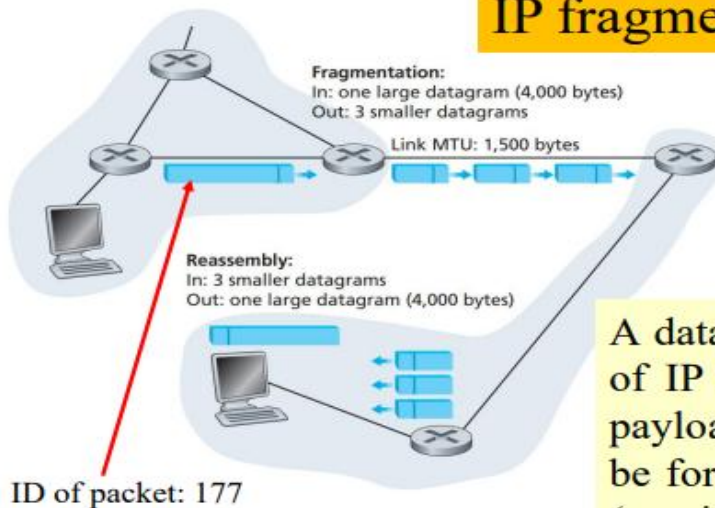
৳ ডিজিটাল ভয়েসের ক্ষেত্রে সঠিকতার (accuracy) চেয়ে দ্রুত ডেলিভারি (fast delivery) বেশি গুরুত্বপূর্ণ।

৳ ফাইল ট্রান্সফারের ক্ষেত্রে দ্রুততার চেয়ে ত্রুটিমুক্ত (error-free) ট্রান্সমিশন বেশি গুরুত্বপূর্ণ।

৳ এই ফিল্ডটি কমিউনিকেশনের তিনটি প্যারামিটারের সাথে সম্পর্কিত: {Delay (বিলম্ব), Throughput (ডেটা পরিবহনের হার), Reliability (নির্ভরযোগ্যতা)}।

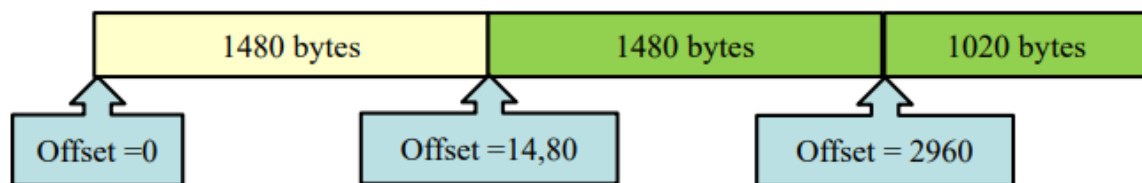
3. Explain operation of IP fragmentation and reassembly with diagram.
Ans: Slide 10 & 11 er table bade sob.

IP fragmentation and reassembly



A datagram of 4,000 bytes (20 bytes of IP header plus 3,980 bytes of IP payload) arrives at a router and must be forwarded to a link with an MTU (maximum transmission unit) of 1,500 bytes.

This implies that the 3,980 data bytes in the original datagram must be allocated to three separate fragments (each of which is also an IP datagram). Suppose that the original datagram is stamped with an identification number of 177.



$$1480 + 1480 + 1029 = 3980$$

11

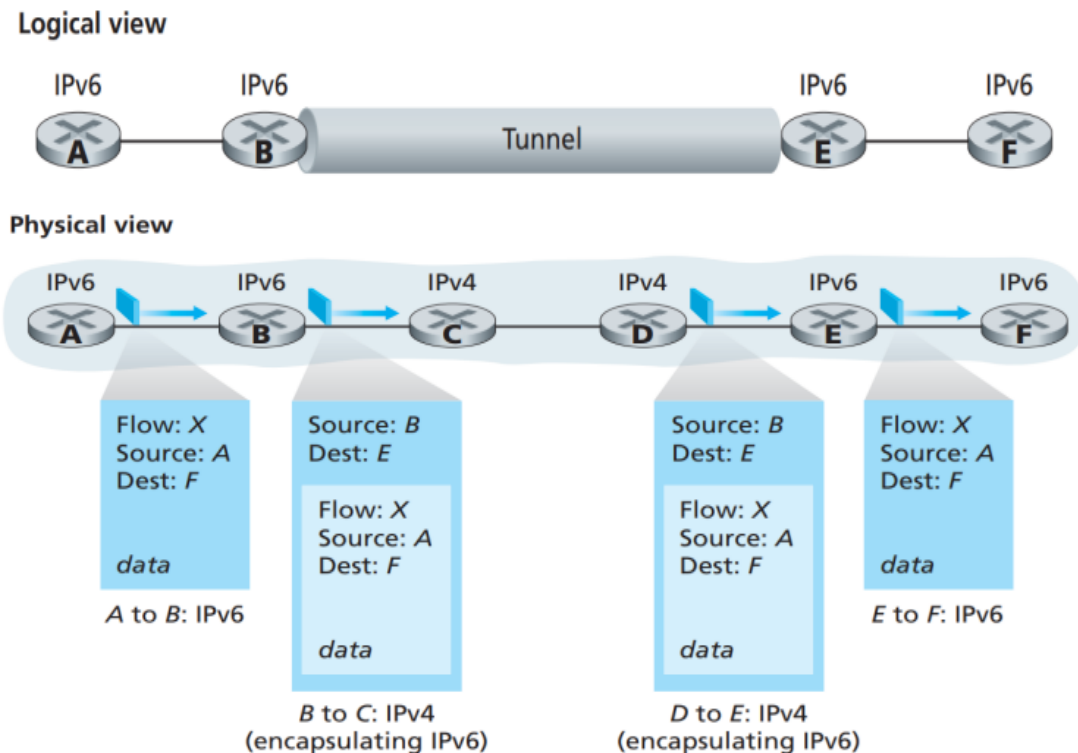
৪,০০০ বাইটের একটি ডাটাগ্রাম (যার মধ্যে ২০ বাইট IP হেডার এবং ৩,৯৮০ বাইট IP payload আছে) একটি রাউটারে পৌঁছায় এবং সেটিকে ১,৫০০ বাইট MTU (Maximum Transmission Unit) বিশিষ্ট একটি লিংকে ফরোয়ার্ড করতে হবে।

এর মানে হলো, মূল ডাটাগ্রামের ৩,৯৮০ বাইট ডেটাকে তিনটি আলাদা ফ্র্যাগমেন্টে (প্রতিটিই একটি করে IP ডাটাগ্রাম) ভাগ করতে হবে।

ধরা যাক, মূল ডাটাগ্রামটিকে ১৭৭ নম্বর আইডেন্টিফিকেশন নম্বর দেওয়া হয়েছে।

4. Give the concept of tunneling with appropriate diagram considering connection of IPV6 routes through IPV4 routes.

Ans: 23 slide diagram only.



5. What is the subnet mask? What is its function.

Ans: Slide 26 or last paragraph.

✓ A **Subnet mask** is a 32-bit number that masks an IP address and divides the IP address into **network address** and **host address**. Subnet Mask is made by setting **network bits** to all '1's and setting **host bits** to all '0's.

IP (class C): 192.168.003.6

Subnet mask: 255.255.255.0

11111111. 11111111. 11111111. 00000000

✓ A subnet mask is used to identify network address of an IP address by performing **bitwise AND operation**.

6. Problems on slide number 34 or Example of 34 slide.

Example-2

If the subnet mask 255.255.240.0 is used for a class B IP address then find the number of subnets and number of hosts/subnet.

Binary: 11111111 . 11111111 . 11110000 . 00000000

Decimal: 255.255.240.0

The number of hosts/subnet = $2^{12} - 2 = 4094$

The number of subnets = $2^4 = 16$

Example-2

If the subnet mask 255.255.255.192 is used for a class C IP address then find the number of subnets and number of hosts/subnet.

The number of hosts/subnet = $2^6 - 2 = 62$

The number of subnets = $2^2 = 4$

Class 9th (Lec-6)

1. Compare datagram and virtual circuit.

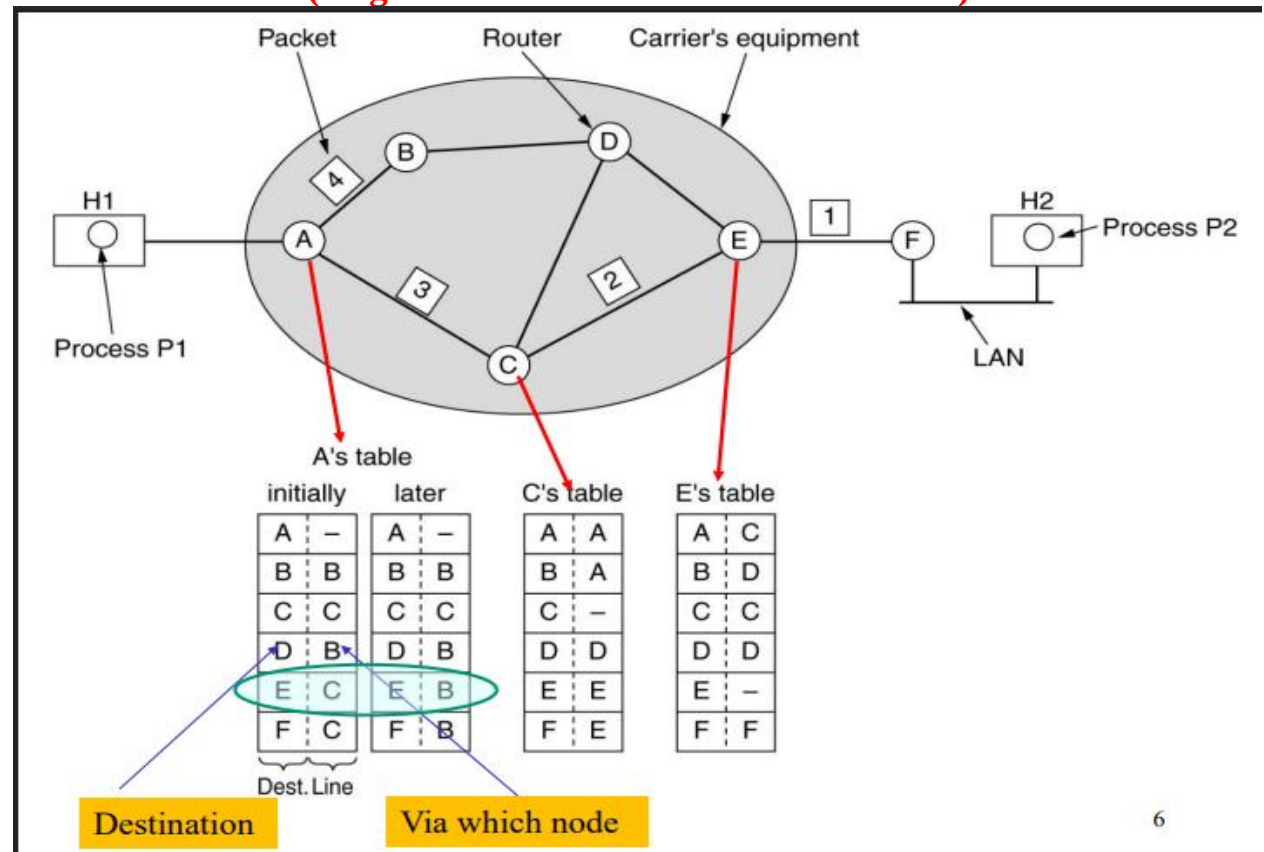
→ Ans: Slide 3

✓ If **connectionless service** is offered, packets are injected onto the subnet individually and routed independently of each other. No advanced setup is needed. In this context, the packets are frequently called **datagram's** (in analogy with telegram) and the network is called a datagram network.

✓ If **connection oriented service** is used, a path from the source to destination router must be established before any data packets can be sent. This connection is called a VC (**virtual circuit**), in analogy with the physical circuits set up by the telephone system, and the network is called a virtual circuit network.

2. Routing table of connectionless service

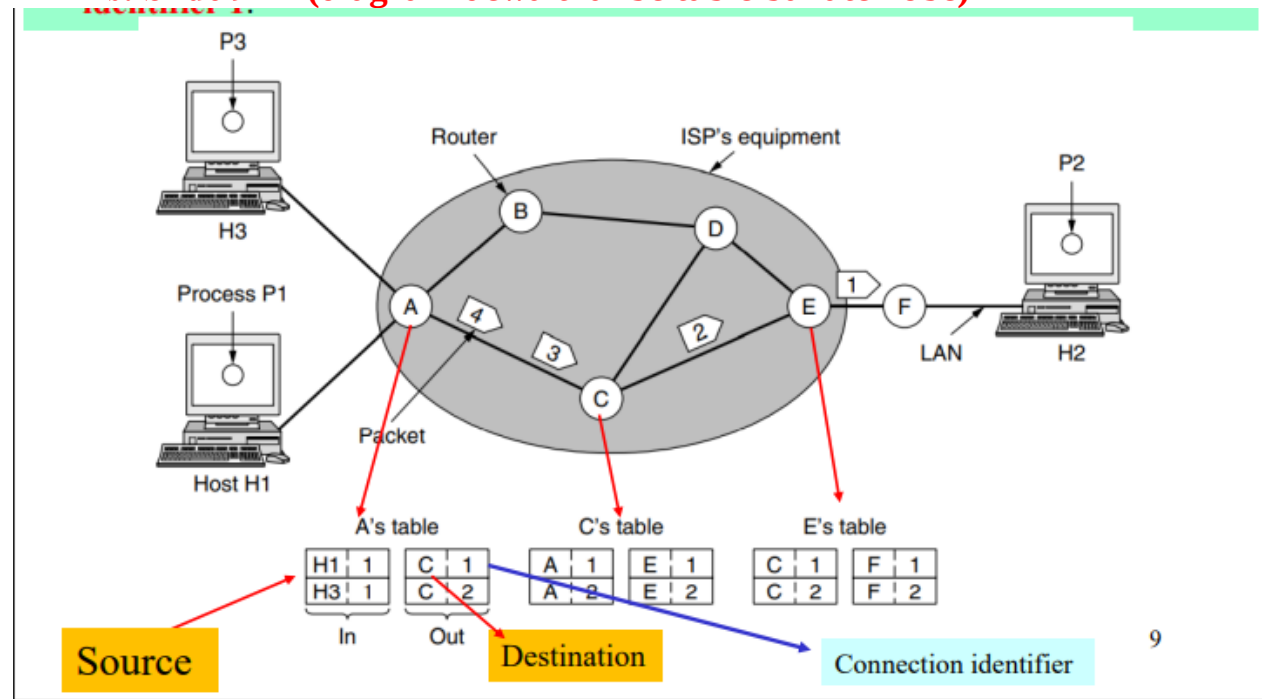
→ Ans: Slide 6 → (diagram dewa thakbe table banate hobe)



6

3. Routing table of connection-oriented service

→ Ans: Slide 9 → (diagram dewa thakbe table banate hobe)



9

4. **Write 3 differences between datagram and virtual circuit**

→ Slide 11

Issue	Datagram subnet	Virtual-circuit subnet
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC ¹¹

5. **Compare non-adapting and adaptive algorithm**

→ Slide 13 (last 2 para)

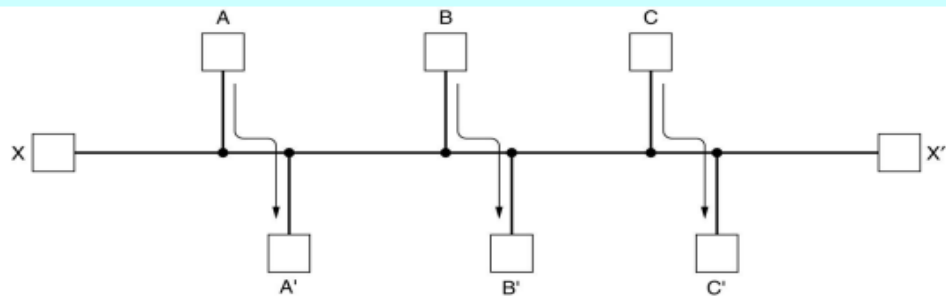
Nonadaptive algorithms (static routing) do not change their routing decisions on any measurements or estimates of the current **topology** and **traffic**.

Adaptive algorithms, in contrast, change their routing decisions to reflect changes in the **topology**, and sometimes changes in the **traffic** as well.

6. **Define fairness and optimality of a network**

→ Ans: Slide 14 (2nd and 3rd para)

- ✓ **Fairness** and **optimally** should be the goal of any engineering problem but sometimes they bring contradictory goals.
- ✓ **Fairness** means equal opportunity to all connected nodes.
- ✓ In context of **optimality**, we actually need to reduce delay of packet. The routing algorithm attempt to minimize the number of hops to reduce delay and increase the throughput i.e. maximize the **efficiency** of the network.



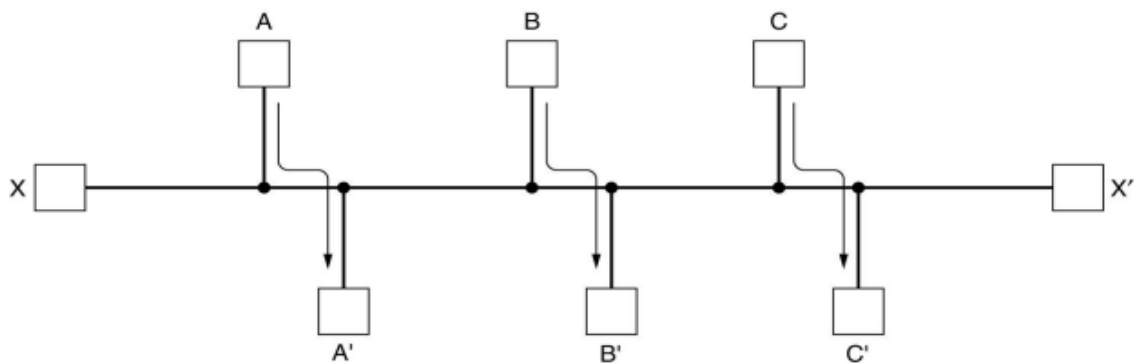
14

7. **How they conflict in practical life.**

→ Ans: Slide 15 → diagram and description

Conflict with efficiency and fairness:

Suppose that there is enough traffic between A and A', between B and B', and between C and C' to saturate the horizontal links. To maximize the total flow, the X and X' traffic should be shut off altogether. Unfortunately, X and X' may not see it that way. Evidently **some compromise** between global **efficiency** and **fairness** to individual connections is needed.

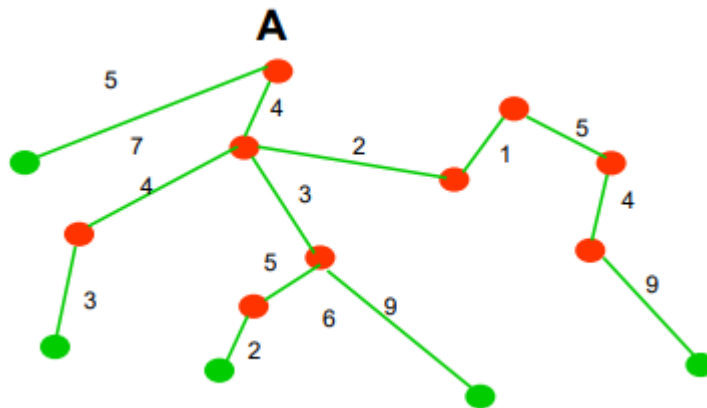
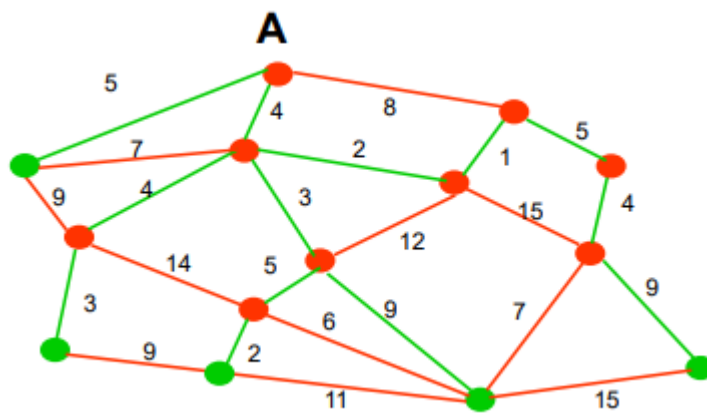
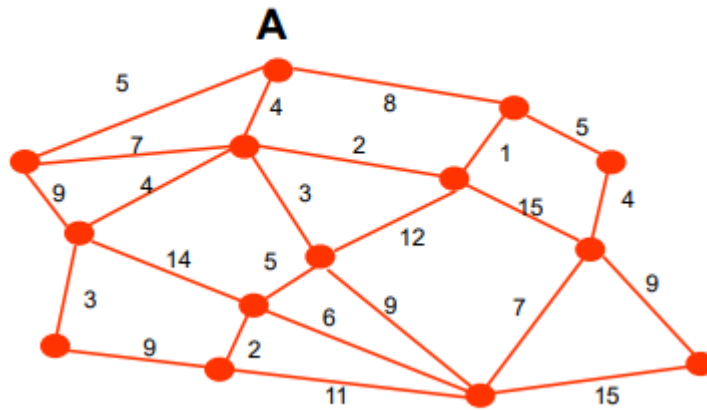


8. **Problem on sink tree**

→ Ans: Slide 18, 19, 20

Exercise

Draw the sink tree taking A as the root/destination.



9. Which parameters are considered to determine path length
→ Slide 21 (6 points).

- ❖ propagation delay,
- ❖ BW/ link capacity in bps,
- ❖ average traffic load on the link,
- ❖ communication cost (in case of leased line),
- ❖ mean queue length of routers along the path (related to delay) etc.
- ❖ The combinations of above criteria the algorithm measures the shortest path.

10. Mention remedial measures of flooding in a network.

→ Slide 33 (last 3 para)

Flooding is a static algorithm in which every incoming packet is sent out on every outgoing line except the one it arrived on like HUB. Flooding obviously generates vast numbers of duplicating packets. Several **remedial measures** are taken against huge duplications of packets:

- Using hop counter in the header of each packet.
- An alternative technique for damming the flood is to keep track of which packets have been flooded, to avoid sending them out a second time.
- Using selective flooding algorithm where the routers do not send every incoming packet out on every line, only on those lines that are going approximately in the right direction.

2nd point Bangla:

বন্যা (flooding) নিয়ন্ত্রণ করার একটি বিকল্প পদ্ধতি হলো—
কোন কোন প্যাকেট ইতিমধ্যে ফ্লাড করা হয়েছে তা ট্র্যাক করে রাখা,
যাতে একই প্যাকেটকে দ্বিতীয়বার আবার পাঠানো না হয়।

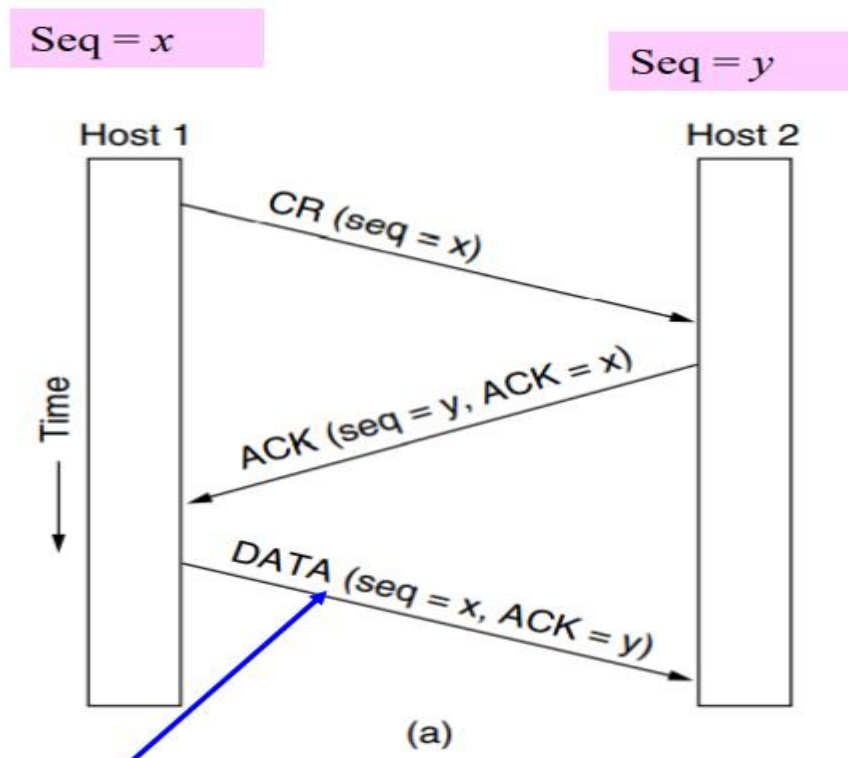
11. **Give the basic concept of logical & physical communication.**

→ Slide-3 (Only text)

The transport layer provides **logical** rather (lot of buffer or memory and wide range of link types between them) than **physical** communication (short circuit no buffer) between application processes.

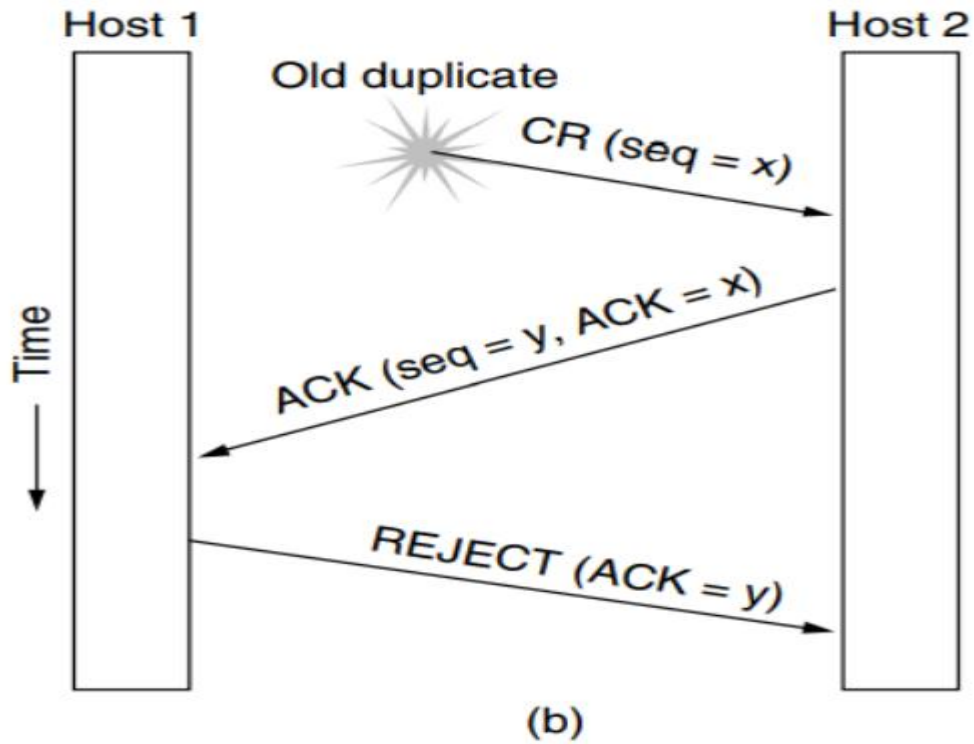
12. **Draw three-way handshaking with appropriate diagram**

→ Slide-11 (Only diagram)

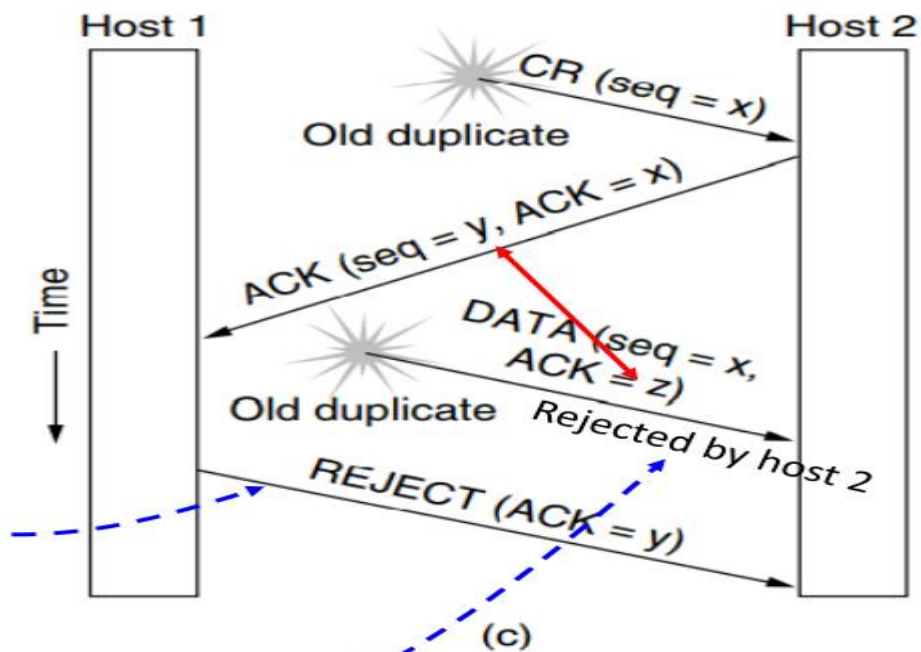


13. **How old duplicate CR is avoided in transport layer**

→ Slide-12 (Only diagram)



14. How both old duplicate CR & old duplicate data are avoided in transport layer
 → Slide-13 (Only diagram)

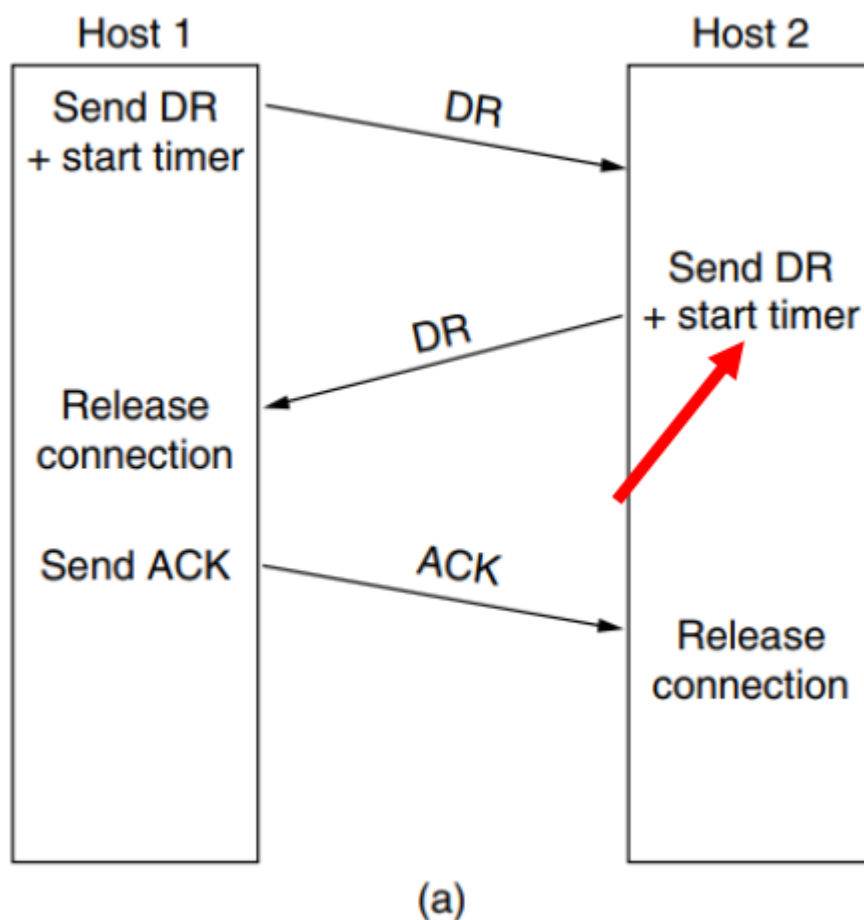


15. Compare asymmetric & symmetric release
 → Slide 14 2nd para.

✓ **Asymmetric release** is the way the telephone system works: when one party hangs up (end the phone call), the connection is broken. **Symmetric release** treats the connection as two separate unidirectional connections and requires each one to be released separately.

16. Draw three-way handshaking for connection release

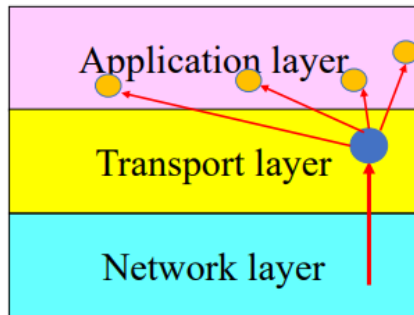
→ Slide-18 (Only diagram)



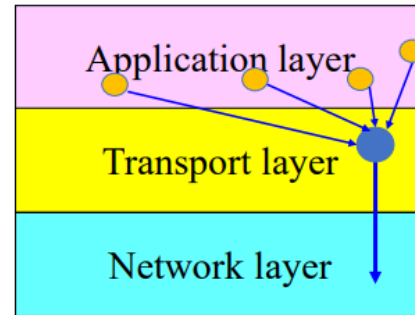
17. Give the basic concept of multiplexing & demultiplexing at transport layer

→ Slide-23 (Full – diagram & text)

- ✓ When the transport layer in a computer receives data packet from the network layer below of it destined to N corresponding processes. Above phenomenon is called **demultiplexing** (S/P converter) at transport layer.
- ✓ Similarly the job of combining data segments from different processes and deliver them to network layer is called **multiplexing** (P/S converter) at transport layer.



Demultiplexing



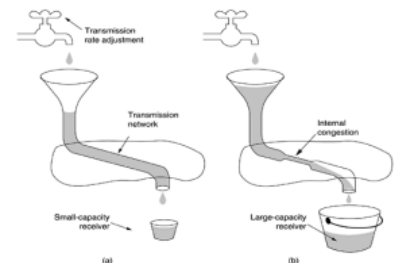
Multiplexing

18. Give the idea of congestion window & receive window

→ Slide 28 or first color portion bade sob.

- ✓ The number of segments to transmit, TCP uses a variable called a **congestion window**, **cwnd**, whose size is controlled by the **congestion situation in the network**.
- ✓ Another variable or window **rwnd** (**receive window**) related to the **limited buffer at the end** (related to buffer flow like sliding window of LLC).
- ✓ The actual size of the window is the minimum of these two.

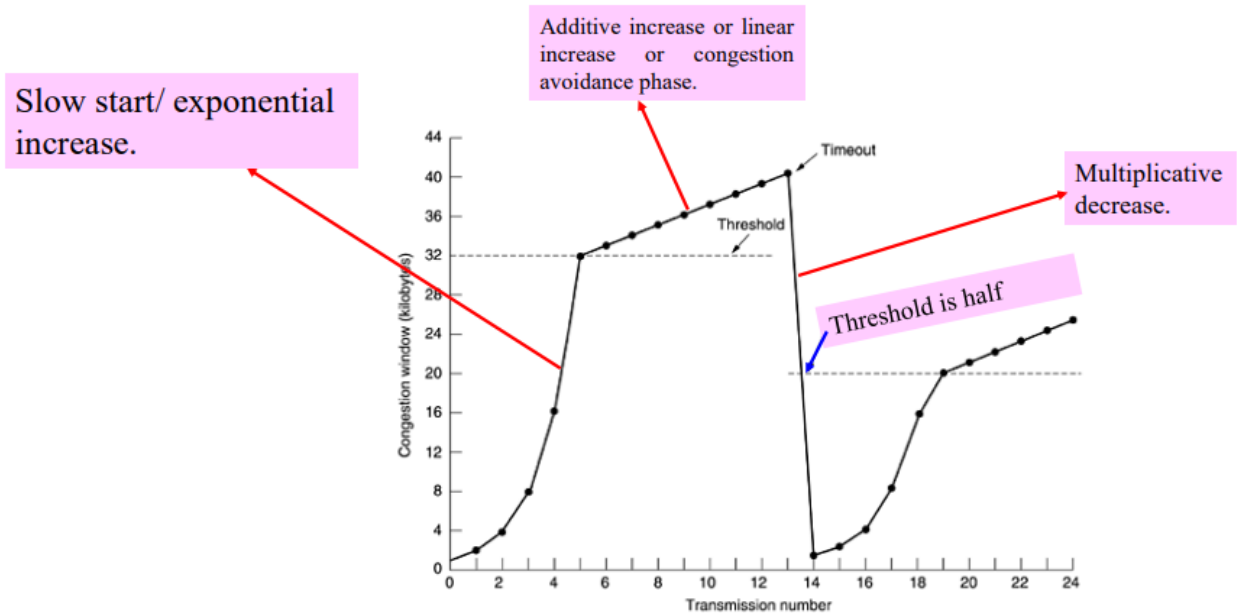
Actual window size = minimum (rwnd, cwnd)



19. Draw the profile of TCP Tahoe

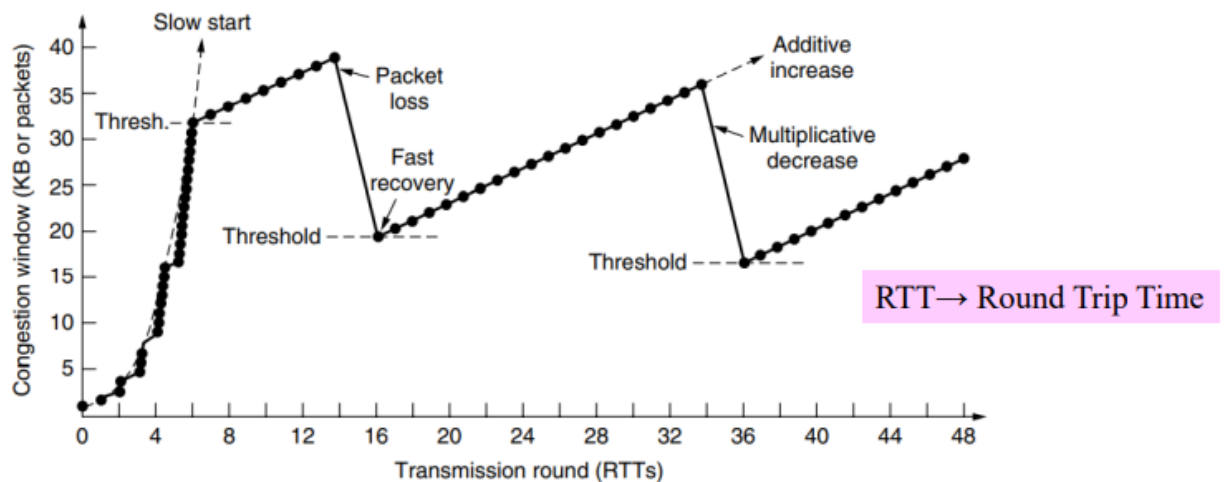
→ Slide-31 (Only diagram)

Internet congestion control algorithm of TCP Tahoe



20. Draw the profile of TCP Reno

→ Slide-34 (Only diagram)



Fast recovery and the sawtooth pattern of TCP Reno

34

Class 11: Date: 3-Apr-2026 Lecture Sheet 8: The Application Layer

Q 1: Compare client-server and P2P architecture at application layer.

Ans: Slide 2 - Para 2 + Slide 3 - Para 1

In a **client-server architecture**, there is an *always-on* host, called the server, which services requests from many other hosts, called clients.

Examples: Web, FTP, Telnet, and e-mail.

✓ In a **P2P architecture**, the application exploits (কাজে লাগান) direct communication between pairs of intermittently (মাঝে মাঝে অথবা অনিয়মিতভাবে) connected hosts, called peers.

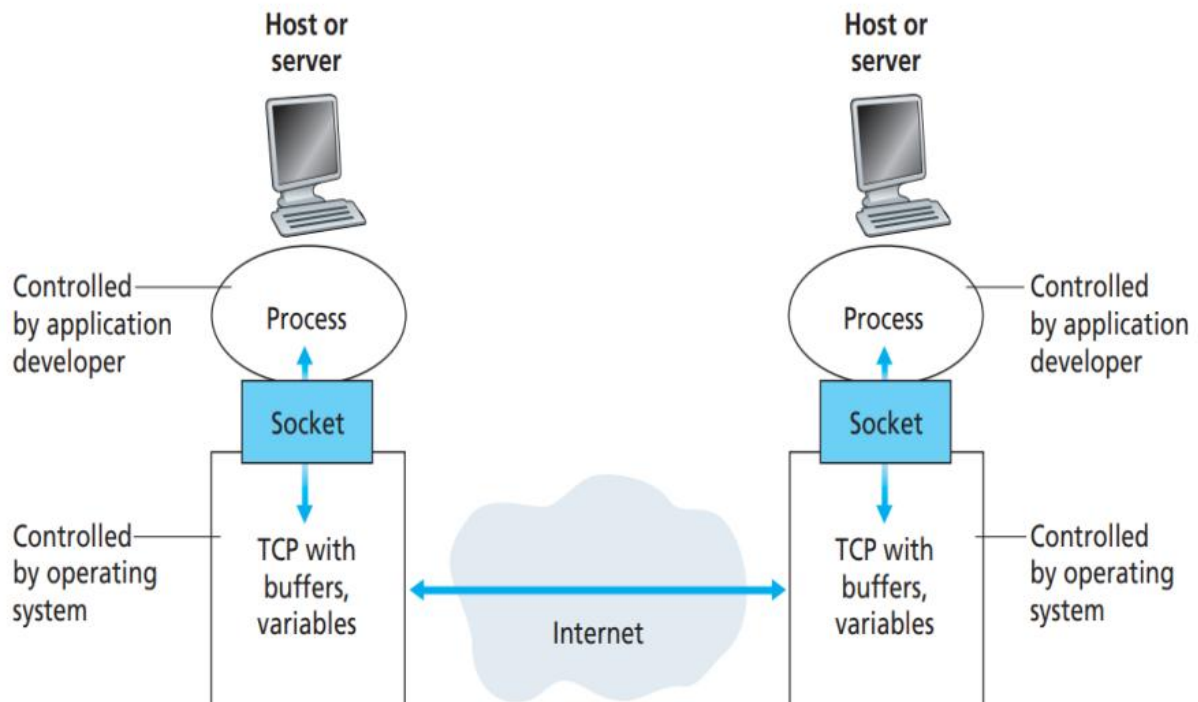
ক্লায়েন্ট-সার্ভার আর্কিটেকচারে, একটি সবসময়-চালু থাকা হোস্ট থাকে, যাকে সার্ভার বলা হয়, যা অনেক অন্যান্য হোস্ট (ক্লায়েন্ট) থেকে আসা অনুরোধগুলো সেবা প্রদান করে।

উদাহরণস্বরূপ: ওয়েব, FTP, টেলনেট, এবং ই-মেইল।

P2P আর্কিটেকচারে, অ্যাপ্লিকেশনটি সরাসরি যোগাযোগকে কাজে লাগায়, যা মাঝেমাঝে (অনিয়মিতভাবে) সংযুক্ত হোস্টগুলোর জোড়ার মধ্যে হয়—এদেরকে **peers** বলা হয়।

Q 2: Draw the arrangement of socket in process to process communication.

Ans: Slide 6 - Only Diagram - No Description.



Q 3: How a process is addressed in a host/network?

Ans: Slide 7- Full Slide

- ✓ A process running on **sending host** needs the full address of the process running on the **receiving host**. To identify each process, two pieces of information is needed : (1) the address of the host (IP address) and (2) the process identifier called port number

- ✓ This port number is needed because in general a host runs many network applications.
- ✓ For example, a Web server is identified by port number 80. A mail server process (using the SMTP protocol) is identified by port number 25.

Q. How a process is addressed in a host.

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Q 4: Give the idea of network application and application protocol.

Ans: Slide 9 - Para 2

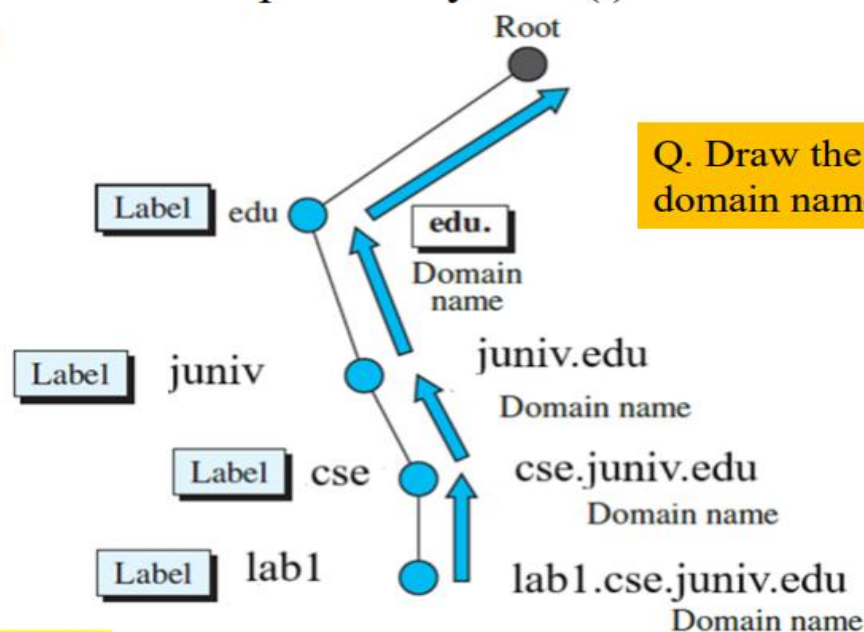
✓ An application layer protocol is associated with each application. Consider World Wide Web browsers that are available (Firefox, Safari, Internet Explorer etc.). You can use any one of these application programs to access a particular site on the Web is because they all use the same application layer protocol: HTTP (Hyper Text Transport Protocol).

প্রতিটি অ্যাপ্লিকেশনের সাথে একটি অ্যাপ্লিকেশন লেয়ার প্রোটোকল যুক্ত থাকে। ধরুন, ওয়ার্ল্ড ওয়াইড ওয়েব ব্রাউজারগুলি যেমন (Firefox, Safari, Internet Explorer ইত্যাদি)। আপনি এই অ্যাপ্লিকেশনগুলির যে কোন একটি ব্যবহার করে নির্দিষ্ট একটি সাইটে প্রবেশ করতে পারেন, কারণ এগুলি সবই একই অ্যাপ্লিকেশন লেয়ার প্রোটোকল ব্যবহার করে: HTTP (হাইপারটেক্সট ট্রান্সফার প্রোটোকল)।

Q 5: Draw the hierarchy of domain name and labels with example.

Ans: Slide 15 - Only Diagram - No Description.

read fr



Q. Draw the hierarchy of domain name and labels.

names are
ne local label

Domain names and labels

Q 6: Mention the function of three phases of POP3.

Ans: Slide 25 – Full

- ✓ During the first phase, **authorization**, the user agent sends a username and a password (in the clear) to authenticate the user.
- ✓ During the second phase, **transaction**, the user agent retrieves messages; also during this phase, the user agent can mark messages for deletion, remove deletion marks, and obtain mail statistics.
- ✓ The third phase, **update**, occurs after the client has issued the quit command, ending the POP3 session; at this time, the mail server deletes the messages that were marked for deletion.

Q. Mention the function of three phases of POP3.

প্রথম ধাপে, **অনুমোদন** (authorization) প্রক্রিয়ায়, ইউজার এজেন্ট একটি ব্যবহারকারীর নাম এবং পাসওয়ার্ড (সরাসরি পাঠানো হয়) প্রেরণ করে ব্যবহারকারীকে প্রমাণীকরণ করতে।

দ্বিতীয় ধাপে, **লেনদেন** (transaction) প্রক্রিয়ায়, ইউজার এজেন্ট মেসেজগুলি উদ্ধার করে; এই ধাপে, ইউজার এজেন্ট মেসেজগুলি মুছে ফেলার জন্য চিহ্নিত করতে পারে, মুছে ফেলার চিহ্ন অপসারণ করতে পারে, এবং মেইল পরিসংখ্যান সংগ্রহ করতে পারে।

তৃতীয় ধাপ, **আপডেট** (update), ঘটে যখন ক্লায়েন্ট "quit" কমান্ডটি জারি করে, যা POP3 সেশন শেষ করে; এই সময়ে, মেইল সার্ভার সেই মেসেজগুলি মুছে ফেলে যেগুলি মুছে ফেলার জন্য চিহ্নিত করা হয়েছিল।